

$$\boxed{\vec{M} = \dot{\vec{L}}}$$

$$\vec{L} = \sum_i m_i \vec{r}_i \times \dot{\vec{r}}_i = \sum_i m_i \vec{r}_i \times (\omega \times \vec{r}_i) = \underline{\underline{J}} \vec{\omega}$$

$$\underline{\underline{J}} = \sum_i m_i (\underline{\underline{1}} r_i^2 - \vec{r}_i \otimes \vec{r}_i) = \int dm (\underline{\underline{1}} r^2 - \vec{r} \otimes \vec{r}), \quad \vec{r} \otimes \vec{r} = \begin{bmatrix} x^2 & xy & xz \\ yx & y^2 & yz \\ zx & zy & z^2 \end{bmatrix}$$

$$\Rightarrow \underline{\underline{J}} = \int dm \begin{bmatrix} y^2+z^2 & -xy & -xz \\ -yx & x^2+z^2 & -yz \\ -zx & -zy & x^2+y^2 \end{bmatrix}$$

$$T = \frac{1}{2} \vec{\omega}^T \underline{\underline{J}} \vec{\omega}$$

in lastrem System S' :

$$\underline{\underline{J}} = \text{diag}(J_x, J_y, J_z); \quad J_x, J_y, J_z \geq 0$$

$$\vec{L} = J_x \omega_x' \hat{i}' + J_y \omega_y' \hat{j}' + J_z \omega_z' \hat{k}'$$

$$\begin{aligned} \dot{\vec{L}} &= J_x (\dot{\omega}_x' \hat{i}' + \vec{\omega}' \times \hat{i}') + \dots \\ &\hookrightarrow \dot{\hat{i}}' = \vec{\omega}' \times \hat{i}' = -\omega_y' \hat{k}' + \omega_z' \hat{j}' \\ \dot{\hat{j}}' &= \omega_x' \hat{k}' - \omega_z' \hat{i}' \\ \dot{\hat{k}}' &= -\omega_x' \hat{j}' + \omega_y' \hat{i}' \end{aligned}$$

$$\rightarrow \vec{M} = \dot{\vec{L}}$$

$$\Rightarrow M_x' = J_x \dot{\omega}_x' - J_y \omega_y' \omega_z' + J_z \omega_z' \omega_y'$$

$$M_y' = J_y \dot{\omega}_y' - (J_z - J_x) (\omega_x' \omega_z')$$

$$M_z' = J_z \dot{\omega}_z' - (J_x - J_y) (\omega_x' \omega_y')$$

① Prosta simetrična vrtačka

$$\vec{M} = 0 \quad \downarrow \quad J = \begin{bmatrix} J & 0 & 0 \\ 0 & J & 0 \\ 0 & 0 & J' \end{bmatrix}$$

$$J_x = J_y = J$$

$$J' = J_z \neq J_x$$

$$\vec{L} = ?$$

$$J\dot{\omega}_x - (J - J')\omega_y\omega_z = 0$$

$$J\dot{\omega}_y - (J' - J)\omega_x\omega_z = 0$$

$$J'\dot{\omega}_z - (J - J')\omega_y\omega_x = 0$$

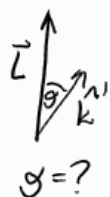
$$J'\dot{\omega}_z = 0 \Rightarrow \omega_z = \omega_0 = \text{konst.}$$

$$J\ddot{\omega}_x = (J - J')\omega_0\dot{\omega}_y = (J - J')\omega_0 \frac{J' - J}{J} \omega_x \omega_0 = -\frac{(J - J')^2}{J} \omega_0^2 \omega_x$$

$$\ddot{\omega}_x = -\underbrace{\frac{(J - J')^2}{J^2} \omega_0^2}_{\Omega_p^2} \omega_x \rightarrow \omega_x = A \cos \Omega_p t + B \sin \Omega_p t$$

$$\omega_y = \frac{J\dot{\omega}_x}{(J - J')\omega_0} = \frac{\dot{\omega}_x}{\Omega_p} = A \sin \Omega_p t - B \cos \Omega_p t$$

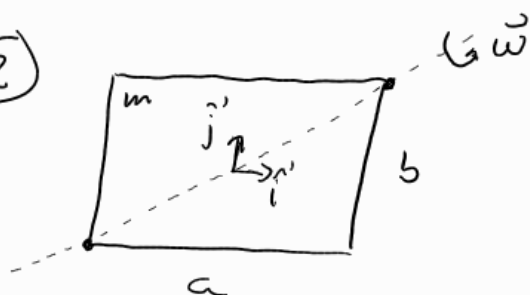
$$\vec{L} = \text{konst. (ker } \vec{M} = 0)$$



$$\varphi = ?$$

$$\vec{L} \cdot \hat{k}' = J' \omega_z' = \text{konst.} \Rightarrow \varphi = \text{konst.} \quad \frac{\vec{L} \cdot \hat{k}'}{|\vec{L}| |\hat{k}'|} = \cos \varphi$$

②



$$J_x = \int (y^2 + z^2) dm =$$

$$= \iint y^2 \frac{m}{5} dx dy = \frac{m}{b} \frac{b^3}{3 \cdot 4} = \frac{mb^2}{12}$$

$$\vec{M} = ? \text{ (u vogačih)}$$

$$J_y = \frac{ma^2}{12}$$

$$J_z = \iint (x^2 + y^2) \frac{m}{ab} dx dy = J_x + J_y = \frac{m}{12} (a^2 + b^2)$$

$$\vec{\omega} = \omega_x' \hat{i}' + \omega_y' \hat{j}' = \frac{a\omega_0}{\sqrt{a^2 + b^2}} \hat{i}' + \frac{b\omega_0}{\sqrt{a^2 + b^2}} \hat{j}'$$

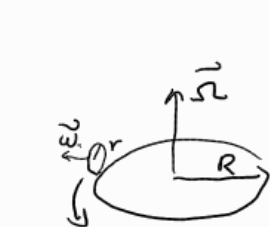
$$M'_x = J_x \dot{\omega}'_x - J_y \omega'_y \omega'_z + J_z \omega'_z \omega'_y = 0$$

$$M'_y = 0$$

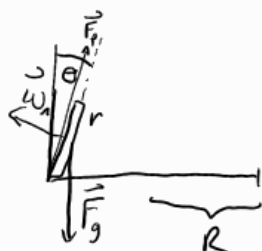
$$M'_z = 0 - J_x \omega'_x \omega'_y + J_y \omega'_x \omega'_y = -\frac{mb^2}{12} \frac{ab\omega_0^2}{a^2+b^2} + \frac{ma^2}{12} \frac{ab\omega_0^2}{a^2+b^2} = \frac{mab\omega_0^2}{12} \frac{a^2-b^2}{a^2+b^2}$$

$$M'_z = rF \underbrace{\sin\varphi}_1 \rightarrow F = \frac{M'_z}{\frac{1}{2}\sqrt{a^2+b^2}} = \frac{mab\omega_0^2}{6} \frac{a^2-b^2}{(a^2+b^2)^{3/2}}$$

Krožec kovanec



r, R, Ω
 $\theta = ?$



$$\vec{F}_g = mg(\cos\theta \hat{j} - \sin\theta \hat{k})$$

$$\vec{F}_p = mg(\cos\theta \hat{j} + \sin\theta \hat{k}) + \underbrace{(R-r\sin\theta)m\Omega^2(\sin\theta \hat{j} - \cos\theta \hat{k})}_{\text{centr. sila (razd. do težišča)}}$$

$$\vec{M}_g = \vec{r} \times \vec{F}_g = 0$$

$$\vec{M}_p = \vec{r}_p \times \vec{F}_p = mg \begin{vmatrix} \hat{i} & \hat{j} & \hat{k} \\ 0 & r & 0 \\ 0 & \cos\theta & \sin\theta \end{vmatrix} + (R-r\sin\theta)m\Omega^2 \begin{vmatrix} \hat{i} & \hat{j} & \hat{k} \\ 0 & r & 0 \\ 0 & \sin\theta & -\cos\theta \end{vmatrix} =$$

$$= mg(-r)\sin\theta \hat{i} + (R-r\sin\theta)m\Omega^2(-r)(-\cos\theta) \hat{i} =$$

$$= [(R-r\sin\theta)m\Omega^2 r \cos\theta - mgr \sin\theta] \hat{i}$$

$$\begin{aligned} J_x &= J_y = \frac{1}{4}mr^2 \\ J_z &= \frac{1}{2}mr^2 \end{aligned}$$

$$\vec{L} = J \vec{\omega}, \quad \vec{\omega} = \vec{\Omega} + \vec{\omega}_i = (0, \Omega \cos\theta, \omega_1 + \Omega \sin\theta)$$

$$\rightarrow \vec{L} = (0, J_y \Omega \cos\theta, J_z (\omega_1 + \Omega \sin\theta))$$

$$-\omega_1 r = \Omega R \quad (\text{lahko izpeljemo iz } v=0 \text{ na stiku})$$

$$\vec{L} = J_y \Omega \cos \theta \hat{j}' + J_z \Omega \left(\sin \theta - \frac{R}{r} \right) \hat{k}' = \begin{vmatrix} \hat{i}' & \hat{j}' & \hat{k}' \\ 0 & \cos \theta & \sin \theta \\ 0 & 1 & 0 \end{vmatrix} = -\Omega \hat{i}' \sin \theta$$

$$= -J_y \Omega^2 \cos \theta \hat{i}' \sin \theta + J_z \Omega^2 \left(\sin \theta - \frac{R}{r} \right) \cos \theta \hat{i}' \quad \begin{vmatrix} \hat{i}' & \hat{j}' & \hat{k}' \\ 0 & \cos \theta & \sin \theta \\ 0 & 0 & 1 \end{vmatrix} = \Omega \cos \theta \hat{i}'$$

$$\vec{L} = \vec{H}:$$

$$(R - r \sin \theta) \mu \Omega^2 \cos \theta - \mu g r \sin \theta = -\frac{1}{4} \mu r^2 \Omega^2 \cos \theta \sin \theta + \frac{1}{2} \mu r^2 \Omega^2 \cos \theta \left(\sin \theta - \frac{R}{r} \right)$$

$$(R - r \sin \theta) \cos \theta - \frac{g}{\Omega^2} \sin \theta = -\frac{1}{4} r \cos \theta \sin \theta + \frac{1}{2} r \cos \theta \left(\sin \theta - \frac{R}{r} \right) \quad / \frac{1}{\cos \theta}$$

$$R - r \sin \theta - \frac{g}{\Omega^2} \tan \theta = -\frac{1}{4} r \sin \theta + \frac{1}{2} r \sin \theta - \frac{1}{2} R$$

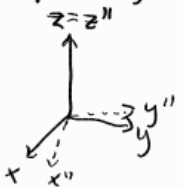
$$\frac{3}{2} R = r \sin \theta + \frac{g}{\Omega^2} \tan \theta - \frac{1}{4} r \sin \theta + \frac{1}{2} r \sin \theta =$$

$$= \frac{5}{4} r \sin \theta + \frac{g}{\Omega^2} \tan \theta = \frac{g}{\Omega^2} \tan \theta \rightarrow \theta = \arctan \left(\frac{3}{2} \frac{R \Omega^2}{g} \right)$$

$r \ll R$

Eulerjevi koti

1) precesija



$$R_\varphi = \begin{bmatrix} \cos \varphi & \sin \varphi & 0 \\ -\sin \varphi & \cos \varphi & 0 \\ 0 & 0 & 1 \end{bmatrix}$$

2) nutacija



$$R_\theta = \begin{bmatrix} 1 & 0 & 0 \\ 0 & \cos \theta & \sin \theta \\ 0 & -\sin \theta & \cos \theta \end{bmatrix}$$

3) vrtenje okoli kretne osi (zasuk)

$$R_\psi = \begin{bmatrix} \cos \psi & \sin \psi & 0 \\ -\sin \psi & \cos \psi & 0 \\ 0 & 0 & 1 \end{bmatrix}$$

$$\vec{\omega} = \dot{\psi} \hat{z} + \dot{\theta} \hat{x}'' + \dot{\varphi} \hat{z}'$$

1) prepisi $\vec{\omega}$ u sistemu $\hat{x}', \hat{y}', \hat{z}'$, T simetrične vrtavke = ?

$$\hat{z}' = R_\psi R_\theta R_\varphi \begin{bmatrix} 0 \\ 0 \\ 1 \end{bmatrix} = R_\psi R_\theta \begin{bmatrix} 0 \\ 0 \\ 1 \end{bmatrix} = R_\psi \begin{bmatrix} \sin \theta \\ \sin \theta \cos \psi \\ \cos \theta \end{bmatrix} = \begin{bmatrix} \sin \theta \sin \psi \\ \sin \theta \cos \psi \\ \cos \theta \end{bmatrix}$$

$$\hat{x}'' = R_\psi R_\theta \begin{bmatrix} 1 \\ 0 \\ 0 \end{bmatrix} = R_\psi \begin{bmatrix} 1 \\ 0 \\ 0 \end{bmatrix} = \begin{bmatrix} \cos \psi \\ \sin \psi \\ 0 \end{bmatrix}$$

$$\vec{\omega}' = \begin{bmatrix} \dot{\psi} \sin \theta \sin \psi + \dot{\theta} \cos \psi \\ \dot{\psi} \sin \theta \cos \psi - \dot{\theta} \sin \psi \\ \dot{\psi} \cos \theta + \dot{\varphi} \end{bmatrix}$$

$$J = \begin{bmatrix} J & 0 & 0 \\ 0 & J & 0 \\ 0 & 0 & J' \end{bmatrix}$$

$$T = \frac{1}{2} \vec{\omega}'^T J \vec{\omega}' = \frac{1}{2} [\omega_x \omega_y \omega_z] \begin{bmatrix} J \omega_x \\ J \omega_y \\ J' \omega_z \end{bmatrix} =$$

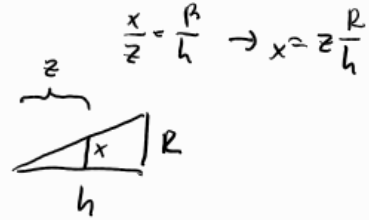
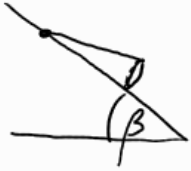
$$= \frac{1}{2} (J \omega_x^2 + J \omega_y^2 + J' \omega_z^2) =$$

$$= \frac{1}{2} (J (\dot{\psi}^2 \sin^2 \theta \sin^2 \psi + \dot{\theta}^2 \cos^2 \psi + 2 \dot{\psi} \dot{\theta} \sin \theta \sin \psi \cos \psi) + J' (\dot{\psi} \cos \theta + \dot{\varphi})^2)$$

$$\begin{aligned}
 & + \dot{\psi}^2 \sin^2 \theta \cos^2 \varphi + \dot{\theta}^2 \sin^2 \varphi - 2 \dot{\psi} \dot{\theta} \sin \theta \sin \varphi \cos \varphi \\
 & + J' (\dot{\psi}^2 \cos^2 \theta + \dot{\psi}^2 + 2 \dot{\psi} \dot{\varphi} \cos \theta) = \\
 & = \frac{1}{2} (J (\dot{\psi}^2 \sin^2 \theta + \dot{\theta}^2) + J' (\dot{\psi}^2 \cos^2 \theta + \dot{\psi}^2 + 2 \dot{\psi} \dot{\varphi} \cos \theta))
 \end{aligned}$$

2. stožec na nagnjenoj podlagi

M, r, h vrh vpet, $\omega = ?$



$$J = \int dm (x^2 + z^2) =$$

$$= \int (x^2 + z^2) \rho dV =$$

$$= \int (r^2 \cos^2 \varphi + z^2) \rho r dr d\varphi dz =$$

$$= \int_0^{2\pi} \int_0^h \left(\frac{R^4}{4h^4} \cos^2 \varphi + z^2 \frac{r^2}{2} \right) dz d\varphi \rho =$$

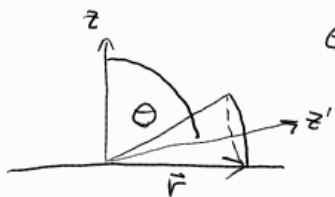
$$= \int_0^{2\pi} \int_0^h \left(\frac{R^4}{4h^4} z^5 \cos^2 \varphi + \frac{R^2}{2h^2} z^3 \right) dz d\varphi \rho =$$

$$= \frac{h^5}{5} \rho \int_0^{2\pi} \left(\frac{R^4}{4h^4} \cos^2 \varphi + \frac{R^2}{2h^2} \right) d\varphi = \frac{h^5}{5} \rho \left[\frac{2\pi R^4}{4h^4} + \frac{R^2}{2h^2} \pi \right] = \frac{\rho \pi}{5} \left[R^2 h^3 + \frac{1}{4} R^4 h \right] =$$

$$J' = \int (x^2 + y^2) \rho dV = \int_0^{2\pi} \int_0^h \int_0^z r^2 \rho r dr d\varphi dz =$$

$$= 2\pi \rho \int_0^h \frac{R^4}{4h^4} z^3 dz = \frac{2\pi \rho R^4}{4h^4} \frac{h^5}{5} = \frac{\pi \rho R^4 h}{10} = \frac{3}{10} M R^2$$

$\beta = 0$



$$\theta = \text{konst.} = \frac{\pi}{2} - \alpha; \tan \alpha = \frac{h}{R}$$

$$\theta = \vec{\omega} \times \vec{r} = \dot{\varphi} \hat{e}_z \times \vec{r} + \dot{\psi} \times \vec{r} = \dot{\varphi} \sqrt{R^2 + h^2} + R \dot{\psi} \rightarrow \dot{\psi} = -\frac{\sqrt{R^2 + h^2}}{R} \dot{\varphi} = -\frac{\dot{\varphi}}{\sin \alpha}$$

$$T = \frac{1}{2} J (\dot{\psi}^2 \sin^2 \theta + \dot{\theta}^2) + \frac{1}{2} J' (\dot{\psi}^2 \cos^2 \theta + \dot{\psi}^2) = \left(\text{prejšnja naloga} \right)$$

$$= \frac{1}{2} J (\dot{\varphi}^2 \cos^2 \alpha) + \frac{1}{2} J' \left(\dot{\varphi}^2 \sin^2 \alpha - \frac{1}{\sin^2 \alpha} \dot{\varphi}^2 \right) = \frac{1}{2} \dot{\varphi}^2 \left(J \sin^2 \left(\frac{\pi}{2} - \alpha \right) + J' \left(\cos^2 \left(\frac{\pi}{2} - \alpha \right) - \frac{1}{\sin^2 \alpha} \right) \right) = \frac{1}{2} J_{\text{ef}} \dot{\varphi}^2$$



$$\vec{z}_T = \frac{\int z dm}{M} = \frac{3}{4} h$$



$$\vec{r}'_T = \vec{z}_T \begin{bmatrix} \cos \alpha \\ 0 \\ \sin \alpha \end{bmatrix}$$

rotacija za φ

$$\vec{r}_T'' = \vec{z}_T \begin{bmatrix} \cos \varphi & -\sin \varphi & 0 \\ \sin \varphi & \cos \varphi & 0 \\ 0 & 0 & 1 \end{bmatrix} \begin{bmatrix} \cos \alpha \\ 0 \\ \sin \alpha \end{bmatrix}$$

$$= \vec{z}_T \begin{bmatrix} \cos \alpha \cos \varphi \\ \cos \alpha \sin \varphi \\ \sin \alpha \end{bmatrix}$$

rot. okoli y
osi, da bo $\begin{bmatrix} 1 \\ 0 \\ 0 \end{bmatrix} \rightarrow \begin{bmatrix} \cos \beta \\ 0 \\ -\sin \beta \end{bmatrix}$

$$\vec{r}_T = \vec{z}_T \begin{bmatrix} \cos \beta & 0 & \sin \beta \\ 0 & 1 & 0 \\ \sin \beta & 0 & \cos \beta \end{bmatrix} \begin{bmatrix} \cos \alpha \cos \varphi \\ \cos \alpha \sin \varphi \\ \sin \alpha \end{bmatrix} =$$

$$= \frac{3}{4} l \begin{bmatrix} \cos \alpha \cos \beta \cos \varphi + \sin \alpha \sin \beta \\ \cos \alpha \sin \varphi \\ -\cos \alpha \sin \beta \cos \varphi + \sin \alpha \cos \beta \end{bmatrix}$$

$$\Rightarrow V = \frac{3}{4} m g h (\sin \alpha \cos \beta - \cos \alpha \sin \beta \cos \varphi)$$

$$\rightarrow L = T - V = \frac{1}{2} J_{\text{eff}} \dot{\varphi}^2 + \frac{3}{4} m g h (\cos \alpha \sin \beta \cos \varphi - \sin \alpha \cos \beta)$$

EL:

$$\varphi: J_{\text{eff}} \ddot{\varphi} + \frac{3}{4} m g h \cos \alpha \sin \beta \sin \varphi = 0$$

za maline kote okrog stabilne lege $\varphi=0$:

$$J_{\text{eff}} \ddot{\varphi} + \underbrace{\frac{3}{4} m g h \cos \alpha \sin \beta}_{\omega^2} \varphi = 0$$

$$\rightarrow \omega^2 = \frac{3}{4} \frac{m g h \cos \alpha \sin \beta}{J_{\text{eff}}} = \frac{5}{4} \frac{h^2 + R^2}{h^2} \frac{m g h \cos \alpha \sin \beta}{m \left(\frac{R^2}{4} + \frac{3}{2} h^2 \right)}$$

$$\begin{aligned} J_{\text{eff}} &= J \cos^2 \alpha + J' \left(\sin \alpha - \frac{1}{\sin \alpha} \right)^2 = J \cos^2 \alpha + J' \left(\frac{\sin^2 \alpha - 1}{\sin \alpha} \right)^2 \\ &= J \cos^2 \alpha + J' (\cos^2 \alpha \cot^2 \alpha) = \cos^2 \alpha (J + J' \cot^2 \alpha) = \\ &= \frac{h^2}{h^2 + R^2} \left(\frac{3}{5} m \left(h^2 + \frac{R^2}{4} \right) + \frac{3}{10} m R^2 \frac{h^2}{R^2} \right) = \\ &= \frac{h^2}{h^2 + R^2} \frac{3}{5} m \left(h^2 + \frac{R^2}{4} + \frac{1}{2} h^2 \right) = \frac{h^2}{h^2 + R^2} \frac{3}{5} m \left(\frac{R^2}{4} + \frac{3}{2} h^2 \right) \end{aligned}$$